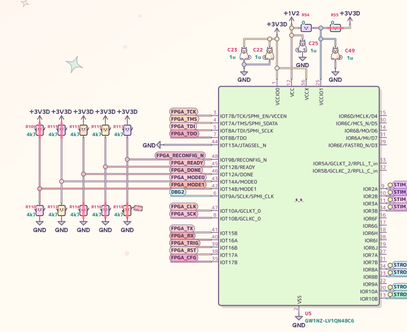
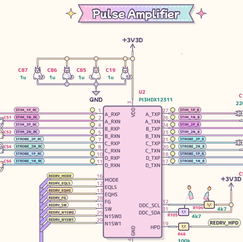
FPGA

Stimulus & Sampling Pulse Timing Generation

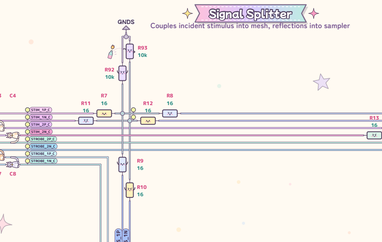


Pulse Amplifier

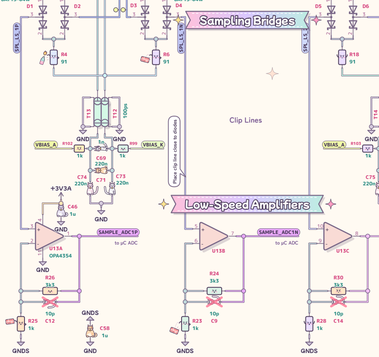


Signal Splitter

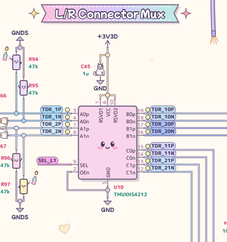
Couples incident stimulus into mesh, reflections into sampler



Sampling Bridges

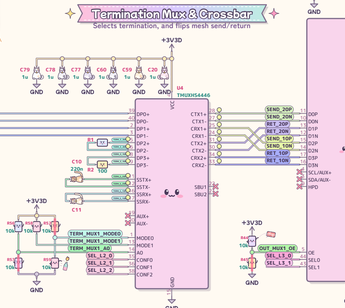


L/R Connector Mux

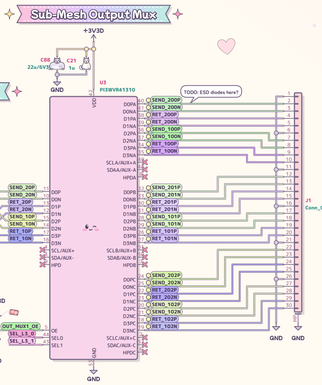


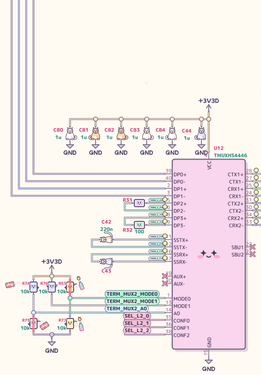
Termination Mux & Crossbar

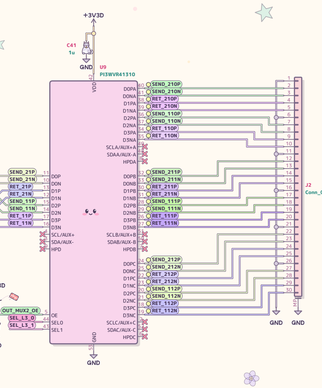
Selects termination, and flips mesh send/return



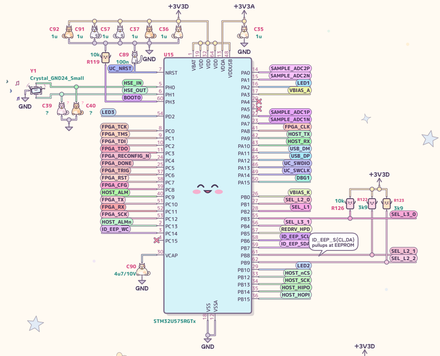
Sub-Mesh Output Mux



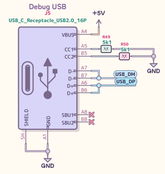


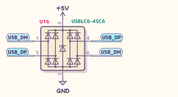


Microcontroller

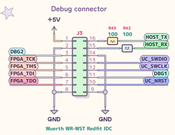


Debug USB

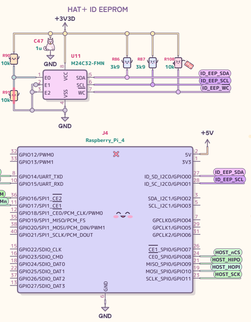




Debug connector

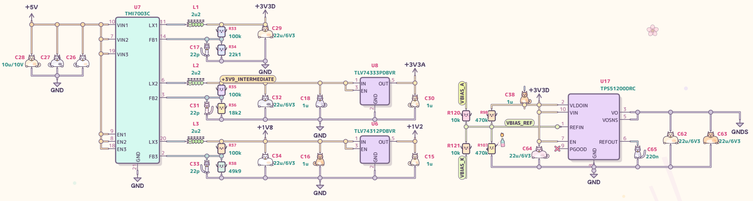


PiHAT+ Connector

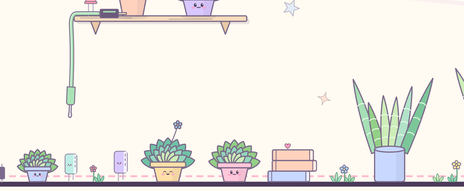


Power Supplies

5V input, outputs: 3V3D (digital), 3V3A (analog, low noise), 1V8 (digital), 1V2 (digital)



Low-Speed Amplifiers



♡ cast of characters ♡



capacitors
more capacitance, more chonk
(the wires go in one end and out the other)

10pF · 1nF · 100nF · 1µF · 4µ7F · 22µF

resistor
grumpy, throws bricks
(and the odd molotov)

inductor
a very hungry caterpillar

crystal
dances at its own frequency ♪



✦ センサー回路図 ✦

yanartas-pillowbox-sensors

~ the pillowbox sensor board ~

sheet ♡ 1

paper A1 · drawn with love ♪